

Addendum v3 Starlifting BH CLEAN FULL

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CVP Fusion Solutions Compendium - Addendum v3

Canonical Equation Update + Starlifting + Black Hole Energy Harvesting (Integration Layer)

Prepared by: Timothy J. Dillon

206 Innovation Inc. Bellevue, Washington

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1. Canonical Equation Update (Locked Notation)

Adopted locked canonical forms (formal documents):

c = local invariant causal speed (relativistic boundary condition)

$c_{\text{eff}} = f(K, R, \nabla K, \partial_t K)$ (curvature-conditioned effective propagation)

$D(c_{\text{eff}}, h, G_{\text{geom}}) = (\delta c_{\text{eff}} / \delta G_{\text{geom}})|_h$ (formal Dillon Operator over gravitational-geometric state)

$\Delta_G c_{\text{eff}} = c_{\text{eff}} - c$ (geometry-induced propagation deviation)

Curvature chain: $T_{\mu\nu} \rightarrow G_{\text{geom}} \rightarrow c_{\text{eff}}$

Notation rule: use G_{geom} (or $\mathcal{G}$); avoid plain G in the Dillon Operator

2. Starlifting Module (Astrophysical Fuel + Energy Infrastructure)

Starlifting is the concept of extracting a tiny fraction of a star's outer plasma (hydrogen/helium) using the star's own energy, guided by controlled heating and magnetic fields. The engineering goal is extremely low extraction rate (e.g., $\sim 0.001\%$ per year) to avoid harming stellar structure while enabling long-term material supply or mega-structure construction.

2.1 Core equations (baseline)

Escape velocity: $v_{\text{esc}} = \sqrt{2GM_{\text{star}} / r_{\text{star}}}$

Specific escape energy: $E_{\text{spec}} = GM_{\text{star}} / r_{\text{star}}$ (J/kg)

Thermal launch energy: $E_{\text{th}} = (1/2)v_{\text{esc}}^2 = GM_{\text{star}} / r_{\text{star}}$

Centrifugal assist: $E_{\text{spec,eff}} \approx E_{\text{spec}} - (1/2)(\omega^2 r_{\text{star}}^2)$

Rotational energy: $E_{\text{rot}} = (1/2)I\omega^2$; $I \approx kM_{\text{star}}r_{\text{star}}^2$ (structure factor k)

2.2 CVP integration (control overlay)

CVP contributes as a control and routing overlay: shaping extraction corridors via curvature-field proxies and EM actuation, tracking stability via Omega convergence metrics, and enforcing TEVR replay for any claimed efficiency improvements. This is a governance/control integration, not a free-energy claim.

3. Black Hole Energy Harvesting Module (Long-Horizon)

Black hole energy harvesting is treated here as long-horizon infrastructure. Candidate mechanisms include accretion conversion, spin-energy extraction in magnetized environments, and extremely long-term Hawking radiation. CVP's role is governance and control: field topology, routing, and stability of extreme electromagnetic environments under TEVR-class auditability.

3.1 Baseline mechanisms (conceptual)

Accretion conversion: $P \approx \eta \cdot \dot{m} \cdot c^2$ (η depends on spin and disk geometry).

Spin-energy extraction: electromagnetic extraction of rotational energy (conceptual; magnetized environment).

Hawking radiation: long-horizon phenomenon; not a practical near-term source.

4. Where this plugs into Fusion